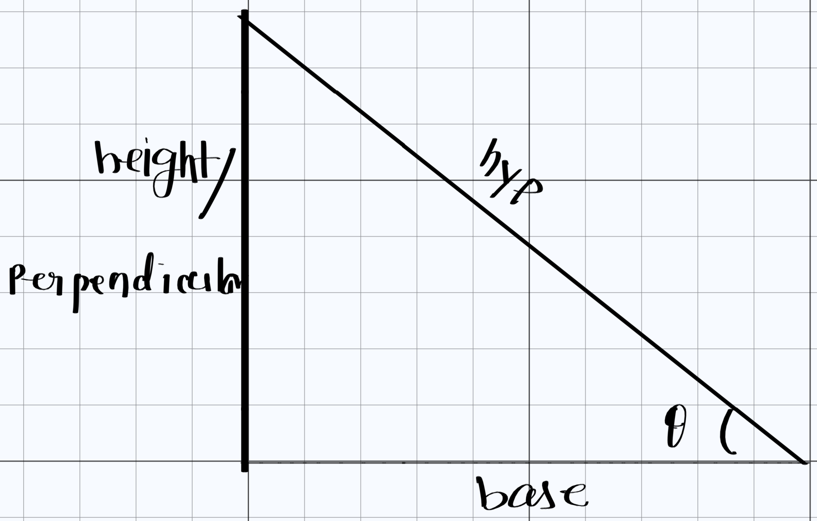
Trigonometry

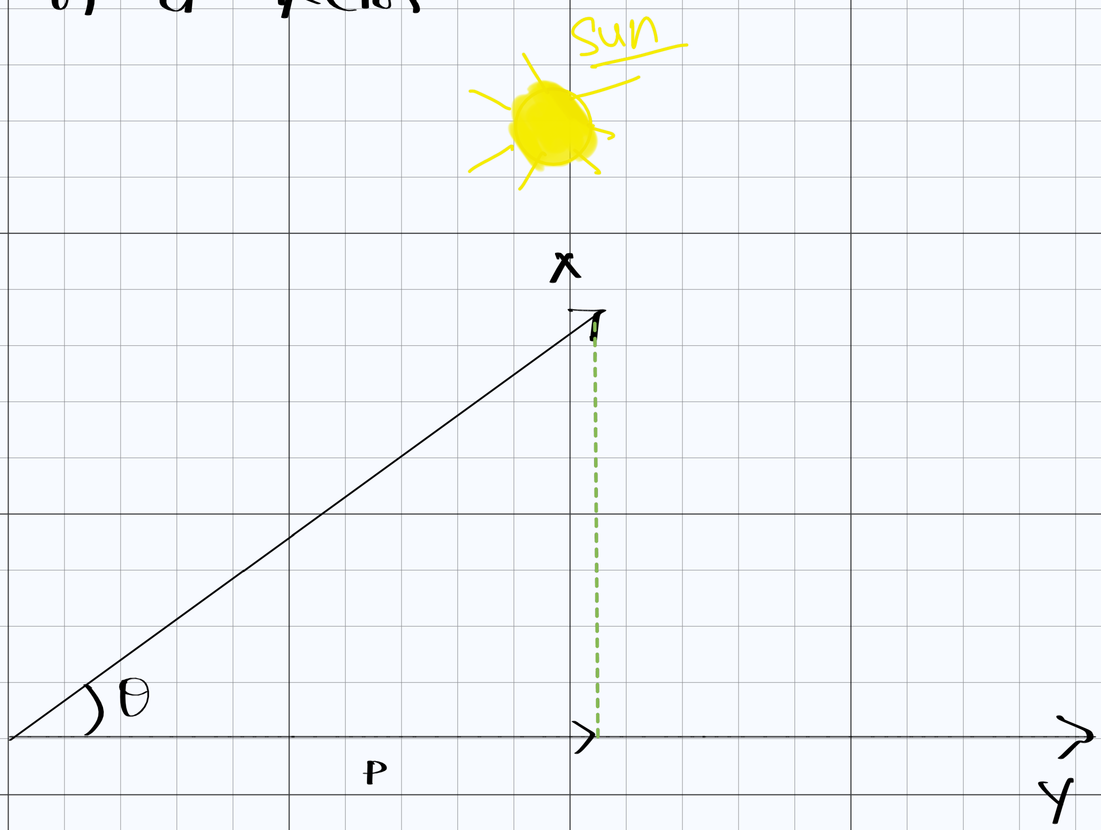


$$\sin\theta = \frac{\text{height}}{\text{hyp}}$$

$$\cos\theta = \frac{\text{base}}{\text{hyp}}$$

$$\tan\theta = \frac{\text{height}}{\text{base}}$$

Projection of a Vector -



$$\cos\theta = \frac{\|p\|}{\|x\|} \longrightarrow (1)$$

angle b/w

$$\cos\theta = \frac{x^T y}{\|x\|\,\|y\|} \rightarrow \text{(II)}$$

$\leftarrow \} \; x \,\&\, y$

from equation (I) & (II)

$$\Rightarrow \frac{\|P\|}{\|x\|} = \frac{x^T y}{\|x\|\,\|y\|}$$

$$\Rightarrow \boxed{\|P\| = \frac{x^T y}{\|y\|}} \leftarrow \text{Scalar projection}$$

$\boxed{\text{Vector projection} = \text{Scalar projection} * \hat{y}}$

$$\Rightarrow \boxed{\frac{x^T y}{\|y\|} * \frac{y}{\|y\|}}$$